Examining Labour Market Shifts Amidst Changing Business Inflation Expectations

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Abstract: A large body of survey evidence shows that firms' subjective macroeconomic expectations diverge from realised conditions and shape pricing and hiring decisions but whether this expectations channel is empirically distinguishable from the effect of realised shocks and whether either of the channel is strong enough to change individual labour mobility, remains largely unexplored in the Indian context. In this paper, I address the following research question: **Do inflationary expectations have an effect on labour mobility and industry switching?** I use the Centre for Monitoring Indian Economy's (CMIE) Industry Outlook for macroeconomic series, which is wave and industry matched with CMIE's Consumer Pyramids Household Survey (CPHS) which tracks same individuals across successive four-monthly waves, and the Business Inflation Expectations Survey (BIES) fielded by the Misra Centre for Financial Markets and Economy at Indian Institute of Management Ahmedabad which surveys firms' backward looking cost assessments and forward looking cost and inflation expectations. These results suggest that for unidirectional industry-switching margin, labour reallocation among Indian workers tracks realised cost shocks more than firms' own expectations of future costs and inflation, a finding which qualifies rather than confirms the more recent macro-expectations literature.

Keywords: Inflation expectations; Labour mobility; Industry switching; Occupational mobility; Adaptive expectations; Firm expectations survey; Household panel data; India; CMIE CPHS; Business Inflation Expectations Survey (BIES); Fixed-effects estimation; Linear probability model; Informalisation; Labour reallocation

1 Introduction

Firms' cost and inflation expectations could plausibly affect workers through atleast four distinct margins: (i) *wage* margin — the pay increments or renegotiated compensation a worker receives while being in the same job; (ii) *migration* margin — the worker has to shift from one place to another with no possibility of role switching in the same company in the new place; (iii) *promotion/demotion* margin — whether a worker's occupational tier or status changes without a change in employer; and (iv) *job-retention and reallocation* margin — whether a worker keeps, looses, or switches out of their current job, employer, or industry altogether. This paper studies a specific part of margin (iv): whether a worker's reported 2-digit NIC industry of employment changes between consecutive time period blocks due to a set of given factors (Macroeconomic variables and BIES variables).

This paper is a correlation analysis, that addresses whether the expectations channel can be empirically distinguishable from the effect of realised macroeconomic conditions and can either channel be strong enough to move individual labour mobility. While there have been experimental analysis in different part of the world (Coibion et al., 2018[4]; Candia et al., 2023[2]), but this paper could be a start to this area of research in the Indian context. This paper mainly asks: *do inflationary expectations have an effect on labour mobility and industry switching?*

I answer this question by combining three rich, high-frequency Indian data sources. The first is the Consumer Pyramids Household Survey (CPHS), a household panel administered by the Centre for Monitoring Indian Economy (CMIE), an individual-level household panel that re-interviews the same respondents roughly every four months (called "Waves"). Each wave records employment status, occupation (NCO), industry (NIC), employment type (formal/informal/casual/self-employed), and a battery of demographic and financial-inclusion variables. The second is CMIE's Industry Outlook database, which supplies wave-matched macroeconomic time series - CPI inflation, WPI input-cost inflation, industry-matched Index of Industrial Production (IIP) growth, and industry-matched bank credit growth - merged onto each CPHS respondent by their reported broad industry of employment (`BROAD_INDUSTRY_NAME`) and survey wave.

The third is the Business Inflation Expectations Survey (BIES), fielded by the Misra Centre at IIM Ahmedabad, which surveys about 8000 manufacturing firms on current cost conditions (Q1–Q3), forward-looking cost expectations (Q4), and CPI inflation expectations (Q5) every month and consolidates the response percentages into expectation groups; BIES responses are aggregated to wave-level weighted means (and, for Q4, a wave-level standard deviation capturing expectation dispersion) and merged onto the CPHS panel by wave. BIES coverage corresponds to CPHS waves 25–39 (approximately January 2022 to April 2026). Merging these data sources, based on NIC 2-digit codes, allows me to test whether individual-level labour reallocation - specifically, whether a worker switches industry or occupational tier between consecutive waves - responds to (a) realised macroeconomic conditions (CPI inflation, WPI input-cost inflation, industry-matched industrial output growth, and industry-matched credit growth), and (b) firms' own elicited expectations of costs and inflation, over and above what is captured by realised conditions alone. This paper addresses only cross-industry reallocation margin of labour mobility and not as a comprehensive measure of every way a worker's family can respond to a shock.

Table 1: Summary Statistics: Industry Switch, Lagged CMIE Macro Variables, and Lagged BIES Variables

	(1) Main Sample			(2) BIES-Manufacturing Sample		
	mean	sd	count	mean	sd	count
=1 if broad industry changed from previous wave	0.349	0.477	353783	0.392	0.488	38473
1-period lag: CPI inflation	5.343	1.158	416343	5.205	1.187	43137
1-period lag: WPI input cost inflation	52.472	73.622	416343	113.671	56.455	43137
1-period lag: industry IIP growth	3.996	0.852	416343	3.834	0.929	43137
1-period lag: Credit industry growth	13.157	5.894	416343	8.894	4.557	43137
AGE_YRS	41.857	11.456	1751291			
Age squared	1883.288	951.299	1751291			
Education level: 0=illiterate, 9=postgraduate	6.624	1.284	1375518			
=1 if currently married	0.782	0.413	1751291			
=1 if member has an active bank account	0.000	0.000	1751291			
=1 if member has formal provident fund coverage	0.000	0.000	1751291			
=1 if member has health insurance protection	0.000	0.000	1751291			
=1 if member has mobile phone access capability	0.000	0.000	1751291			
BIES Q1: profit-margin net balance vs. normal, 1-wave lag (pp)				-0.412	0.164	187564
BIES Q2: sales-level net balance vs. normal, 1-wave lag (pp)				-0.404	0.049	187564
BIES Q3: backward-looking cost change, weighted mean, 1-wave lag (%)				5.208	1.180	187564
BIES Q4: forward-looking cost expectation, weighted mean, 1-wave lag (%)				4.752	0.613	187564
BIES Q5: CPI inflation expectation, weighted mean, 1-wave lag (%)				4.712	0.517	187564
Observations	1751291			187564		

Main Sample: all earning members observed in >= 2 waves (used in Tables 2-3). BIES-Manufacturing Sample: manufacturing workers (NIC 10-33) in waves with BIES coverage, earning members observed in ¿=2 waves (used in Table 4). Sample period: CPHS waves 25-37 (approx. January 2022 - April 2026), the BIES coverage window; CMIE macro series and CPHS employment records are matched by wave. Individual controls (estimated, not reported): AGE_YRS (age, years), AGE2 (age squared), EDU_NUM (education, 0=illiterate to 9=postgraduate), MARRIED (currently married), BANK_ACC (has bank account), PF_COV (provident fund coverage), HEALTH_INS (health insurance), MOBILE_ACC (mobile phone access), RELATION_NUM (relationship-to-household-head dummies), WORK_PLACE_NUM (place-of-work dummies). Source: CMIE CPHS.

The Main Sample (Section 4) consists of all CPHS respondents who are earning members observed in at least two consecutive waves — the minimum required both for the individual fixed-effects estimator and for the industry-switch outcome to be defined. This yields 1,751,291 person-wave observations for the demographic controls; restricting further to observations with non-missing lagged macro regressors leaves 416,343 (Table 1, column (1)).

The BIES-Manufacturing Sample (Sections 5.1–5.2) further restricts the Main Sample to (i) waves with BIES coverage and (ii) manufacturing workers, defined as 2-digit NIC divisions 10–33. Applying the same two-sample logic as above, the BIES-Manufacturing Sample yields 43,137 person-wave observations with non-missing lagged CMIE macro proxies, of which 38,473 also have a non-missing industry-switch outcome (Table 1, column (2)).

Table 1 reports summary statistics for the two samples side by side. On average, 34.9 % of Main-Sample person-wave pairs involve an industry switch, compared with a higher 39.2 % in the BIES-Manufacturing Sample — consistent with greater job-to-job churn among manufacturing workers relative to the broader CPHS population. The BIES-Manufacturing subsample also faces somewhat higher mean WPI input-cost inflation (113.7 vs. 52.5 in Main-Sample index units) and lower mean credit growth (8.9 % vs. 13.2 %), reflecting both its shorter (BIES-covered) time window and its manufacturing-specific composition. Demographic and financial-inclusion controls (age, education, marital status, bank account, PF coverage, health insurance, mobile access) are summarized only for the Main Sample in Table 1; in the BIES-Manufacturing regressions of Table 5 these enter as unreported controls rather than headline variables of interest.

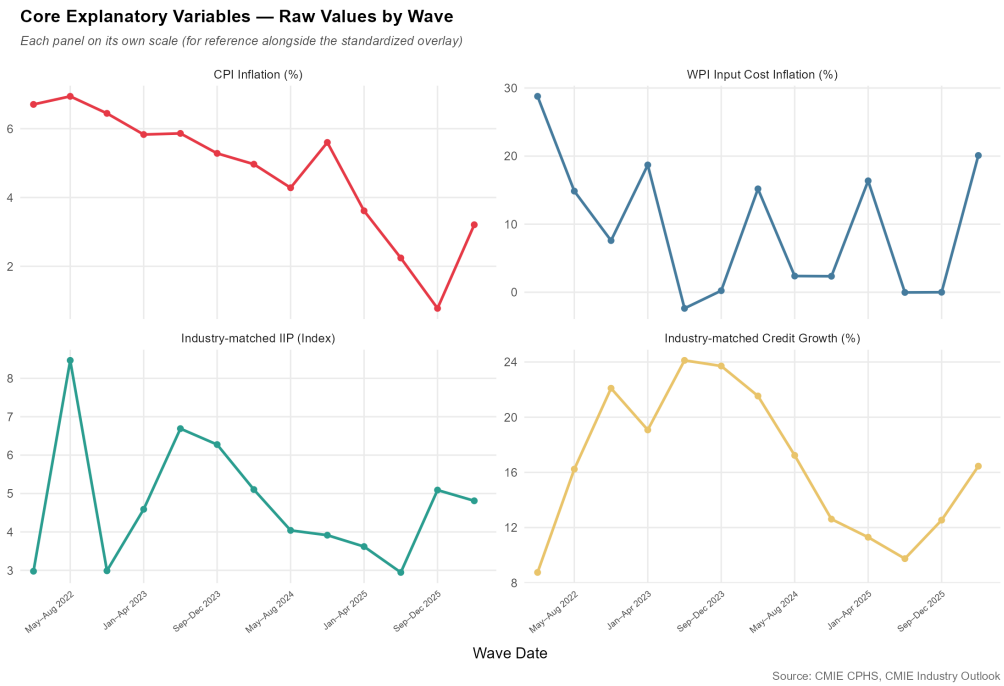


Figure 1: Wave-level means of the four Set A explanatory variables — CPI inflation, WPI input-cost inflation, industry-matched IIP growth, and industry-matched credit growth — over CPHS waves 25–39 (January 2022–April 2026), the BIES coverage window. Each panel is plotted on its own scale in raw units to preserve the true magnitude and volatility of each series. Source: CMIE Consumer Pyramids Household Survey (CPHS), CMIE Industry Outlook.

Figure 1 plots the raw wave-level values of the four core macro regressors from the first BIES-covered wave (Jan–Apr 2022) through Jan–Apr 2026. CPI inflation declines fairly steadily from above 6% in early 2022 to below 1% by late 2025, before rebounding in the most recent wave; WPI input-cost inflation and industry-matched credit growth are considerably more volatile wave-to-wave, while industry-matched IIP growth shows two pronounced spikes (mid-2022 and mid-2023) followed by a gradual decline.

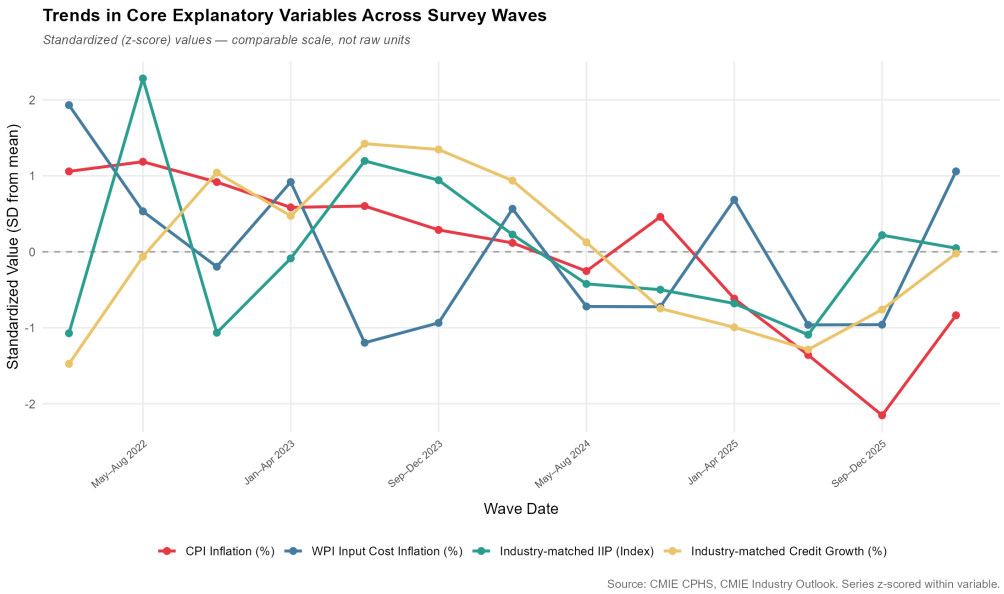


Figure 2: The same four series as Figure 1, standardized to mean zero and unit standard deviation within each variable (z-scored) and overlaid on a common axis, to allow direct visual comparison of relative co-movement and timing across series that differ substantially in units, scale, and volatility. Source: CMIE CPHS, CMIE Industry Outlook. Series z-scored within variable; dashed line marks the sample mean ($z = 0$).

Because the four series are measured in different units and have very different scales and volatilities, Figure 2 re-plots them jointly after standardizing each series to mean zero, unit standard deviation, which makes it possible to compare their relative co-movement on a single axis. The standardized series show industry-matched credit growth and IIP growth tracking each other closely through 2023 before diverging, while CPI inflation traces a visibly smoother, more persistent decline than the other three series — consistent with its lower wave-to-wave volatility relative to WPI input costs shown in raw units in Figure 2. This co-movement pattern motivates treating aggregate business-cycle variation as a confound to be absorbed by wave fixed effects in the two-way specification (Equation 2).

The empirical strategy proceeds in two complementary tracks. The main-sample track uses the full CPHS panel merged with CMIE Industry Outlook macroeconomic series, estimating individual and industry fixed-effects linear probability models (LPMs) of industry switching on one-wave-lagged macro variables, following an adaptive-expectations logic in which lagged realised values proxy for the macroeconomic conditions that agents could plausibly have anticipated. The robustness-check track restricts the sample to CPHS manufacturing workers (NIC divisions 10–33) observed in waves with BIES coverage, and directly compares the explanatory power of lagged CMIE macro proxies against lagged BIES-elicited firm expectations for the same outcome and the same workers, allowing

a head-to-head test of whether *directly elicited* expectations carry independent information beyond realised conditions. Akarsu et al. (2025)[1], discussed further in 2.1, provide a complementary, experimentally-identified estimate of how an induced revision in a firm's own expectations moves the same firm's pricing and hiring in Turkey; the exercise here asks a related but distinctive question with representative, non-experimental data: do firms' *equilibrium* expectations, as they naturally arise are elicited in BIES, leave a detectable footprint on realised worker mobilility, over and above realised macro conditions, in a labour market where informal and causal employment - rather than formal hiring and firing is the dominant margin of adjustment?

The Main Sample results, however, provide only a limited evidence that realised macro conditions predict industry switching once aggregate wave effects are absorbed, and a leads-and-lags placebo test further shows that current switching is also predicted by *future* realised macro conditions - pattern consistent with clean causal, backward looking reading of the lagged coefficients. The sharper test comparing lagged CMIE macro proxies against lagged BIES-elicited firm expectations on the identical manufacturing subsample and wave set shows that realised WPI input-cost inflation significantly predicts industry switching, while none of the BIES expectations do despite a larger estimation sample, and this result is the same for the placebo test as well. Taken together, the evidence suggests that, for the industry-switching margin studied here, Indian workers' reallocation tracks *realised* cost shocks more closely than it tracks firms' own *elicited* expectations of future costs and inflation. This qualifies, rather than confirms, the presumption in much of the macro-expectations literature that subjective expectations are a first order driver of real labour-market decisions, and points instead to backward-looking, liquidity and survival driven adjustment as the dominant force behind labour reallocation in a labour market where informal and casual employment absorbs most of the shock. This matters for how policymakers and firms should think about the transmission of inflation surprises to Indian labour markets like interventions or communication strategies aimed at managing firms' or workers' *expectations* of future costs may have limited traction to actual reallocation behaviour relative to interventions that address the *realised* cost and liquidity conditions driving it.

2 Literature Review

This paper's central contribution is an observational, representative-sample test of whether Indian workers' industry-switching behaviour tracks realised macroeconomic shocks or firms' directly elicited expectations of future costs and inflation, exploiting a novel merge of a firm expectation survey (BIES) onto a large household labour panel (CPHS). This research draws on, and contributes to, three strands of literature: (i) the literature on how firms and households form macroeconomic expectations and how those expectations feed into real decisions; (ii) the macro-labour literature on how aggregate shocks and labour-market institutions jointly determine employment and reallocation outcomes; and (iii) the emerging literature on informal labour markets and job-loss dynamics in India.

2.1 Firm and household macroeconomic expectations

A large body of work has documented that the expectations that firms and households report in surveys deviate systematically from full-information rational expectations, and that these deviations have real consequences. Coibion et al. (2018)[4] provide foundational evidence, that firm expectations about inflation and other macro aggregates are poorly anchored, exhibit substantial disagreement,

and rational inattention. Candia et al. (2023)[2] synthesise this literature in expectations correlating with firms' pricing, investment, and hiring decisions.

Most closely related to the present paper is Akarsu et al. (2025)[1], who use a randomised control trial embedded in a firm survey in Turkey to show that experimentally induced revision in a firm's inflation expectations causally moves from the same firm's own subsequent pricing and hiring decisions. My setting does not have experimental variation in expectations; instead, I exploit the panel structure of a firm expectations survey (BIES) merged with a household labour-market panel (CPHS) to ask a complementary, non-causal but nonetheless informative question: do the correlations between labour reallocation and macro variables look different, in sign or magnitude, when the labour-market-relevant "expectations" regressor is a firm's own elicited probabilistic forecast rather than an aggregate statistic compiled by a government or research agency? A null result on the BIES regressors is therefore not evidence against Akarsu et al.'s (2025)[1] causal finding; at most it shows that whatever local causal effect elicited expectations have on an individual firm's own decisions, that effect is not detectable as an aggregate, wave-level predictor of realised work reallocation in the Indian data used here - a plausible reason for this divergence is elaborated in Section 2.3.

Gautier et al. (2025)[5] study firms' inflation and wage expectations show that firms revised their wage-setting expectations upward in response to elevated inflation expectations, with heterogeneous pass-through across sectors depending on labour-market tightness. Their finding that wage and price expectations co-move, and that this co-movement varies by sector-level labour-market conditions, motivates my choice to interact macro shocks with sectoral and occupational classifications (NIC and NCO) in the labour-reallocation regressions below, and to conduct separate heterogeneity analysis by gender and rural/urban location, given well-documented differences in labour-market tightness and bargaining power across these groups in the Indian context.

2.2 Macroeconomic shocks, labour-market institutions, and Reallocation

The classical macro-labour literature emphasises that the employment response to macroeconomic shocks depends jointly on the shock itself and on labour-market institutions. Chen et al. (2003)[3] develop a dynamic labour-demand model in which the effect of firing costs on employment is expected to fall, higher firing costs primarily discourage hiring rather than deterring layoffs, so that institutions that are protective in good times can be harmful to employment exactly when protection seems most needed. This motivates both the lagged/adaptive-expectations specifications (which allow effects to build up over one to two waves) and the heterogeneity analysis by region and gender.

Goux et al. (2001)[6] study the dynamics of labour demand under fixed-term contracts using French establishment-level panel data and GMM estimators (Hansen, 1982[7]), showing that the elasticity of labour demand to output and cost shocks differs sharply between fixed-term and permanent contract margins. Their fixed-effects and lag-based identification strategy parallels the individual fixed-effects approach, and underscores the importance of distinguishing margins of adjustment in a setting - like India's - where formal job protection covers only a minority of workers.

2.3 Informality and labour-market risk in India

Kumari et al. (2025)[8] study how Indian workers cope with labour-market risk following formal job loss, documenting substantial and persistent earnings losses and a marked shift toward informal employment following displacement from formal jobs. Their informalisation finding that adverse shocks push workers from formal into informal, casual, or lower-tier work rather than simply raising unemployment - is the central reason to expect the Indian expectations channel might look different from Turkey's: with roughly four-fifths of Indian workers informally employed and largely outside the Industrial Disputes Act's job-protection regime, the margin of adjustment available to Indian firms facing a cost or demand shock is not primarily the formal hire/fire margin, but movement of workers across formal, informal, casual, and self-employed tiers. If reallocation for the median Indian worker is driven by contemporaneous liquidity and survival constraints rather than forward-looking job search, realised shocks should dominate elicited expectations in predicting industry switching even if firms' expectations-formation process is qualitatively similar to that documented in Turkey - the pattern the results below in fact show. This also motivates the financial-inclusion controls (bank account access, provident fund coverage, health insurance, mobile access) included throughout as buffers against distress-driven reallocation.

3 Methodology and Empirical Strategy

Taken together, this literature motivates two testable implications for industry-switching margin: (i) macro shocks operating through firm cost and margin channels (WPI input-cost inflation) and through household purchasing power (CPI inflation) should predict industry switching, with the CPI effect's sign left ambiguous a priori since it can trigger both voluntary search and forced churn; and (ii) if firms' own directly elicited expectations carry information beyond realised conditions, BIES regressors should retain explanatory power net of the CMIE proxies in the manufacturing sub-sample.

The empirical analysis in this paper tests two parallel sets of hypotheses, both stated in terms of the single outcome variable actually estimated throughout the paper: `INDUSTRY_SWITCH`, a binary indicator for whether a worker's 2-digit NIC industry changes between consecutive waves. **Set A** tests realised macroeconomic conditions (the transmission channel actually operating in the economy); **Set B** tests firms' directly elicited (BIES) expectations of costs and inflation (the channel implied by the research question itself). Comparing the two sets tests whether industry switching responds to *actual* conditions, *anticipated* conditions, or both.

Because `INDUSTRY_SWITCH` does not distinguish whether a switch moves a worker up or down the occupational or formality ladder, every hypothesis below is stated only in terms of the *probability of switching*, not its direction. Directional hypotheses about upward versus downward occupational mobility - which would require the `UPWARD_MOB`/`DOWNWARD_MOB` outcomes defined in the Appendix (Table 7) - are a natural extension of this framework but are not estimated in this paper, and so are deliberately omitted here rather than posed and left untested.

3.1 Baseline specification: Realised macro conditions (Set A)

The baseline specification estimates a linear probability model (LPM) of industry switching on one-wave-lagged realised macroeconomic conditions, following the adaptive-expectations logic described in the Appendix (Section 7.5): since firm- and worker-level expectations are not observed at the

individual-wave frequency for the full CPHS sample, lagged realised macro values proxy for the macroeconomic conditions that agents could plausibly have anticipated going into the current wave. The one-way fixed-effects specification is

$$\text{INDUSTRY_SWITCH}_{i,t} = \beta_1\,\text{L_INF_CPI}_{i,t-1} + \beta_2\,\text{L_WPI_INPUT}_{i,t-1} + \beta_3\,\text{L_IIP_INDUSTRY}_{i,t-1} + \beta_4\,\text{L_CREDIT_INDUSTRY}_{i,t-1} + X'_{i,t}\gamma + \delta_{j(i)} + \delta_{j(i)} \times \text{WAVE_NO} + \alpha_i + \varepsilon_{i,t} \tag{1}$$

where i indexes individuals, t indexes survey waves, $j(i)$ denotes individual i's broad industry of employment, α_i is an individual fixed effect (absorbing all time-invariant individual heterogeneity), $\delta_{j(i)}$ is a broad-industry fixed effect, $\delta_{j(i)} \times \text{WAVE_NO}$ is an industry-specific linear wave trend, and $X_{i,t}$ is the vector of individual controls (age, age-squared, education, marital status, bank-account access). The two-way fixed-effects specification augments Equation (1) with wave fixed effects λ_t:

$$\text{INDUSTRY_SWITCH}_{i,t} = \beta_1\,\text{L_INF_CPI}_{i,t-1} + \beta_2\,\text{L_WPI_INPUT}_{i,t-1} + \beta_3\,\text{L_IIP_INDUSTRY}_{i,t-1} + \beta_4\,\text{L_CREDIT_INDUSTRY}_{i,t-1} + X'_{i,t}\gamma + \delta_{j(i)} + \delta_{j(i)} \times \text{WAVE_NO} + \lambda_t + \alpha_i + \varepsilon_{i,t} \tag{2}$$

Wave fixed effects absorb any aggregate, economy-wide shock common to all individuals in a given wave, so identification of β_1–β_4 in Equation (2) comes purely from *cross-industry* variation in macro conditions within a wave, net of the industry-specific linear trend. Both specifications are estimated by the within (fixed-effects) estimator, with standard errors clustered at the individual level (`vce(cluster MEM_PANEL_ID)`) throughout, and coefficients on the (unreported) industry fixed effects and industry-wave trends are absorbed but not tabulated.

3.2 BIES robustness specification: direct expectations (Set B)

To test whether firms' *directly elicited* expectations carry independent explanatory power beyond realised conditions, I re-estimate the industry-switching model on the BIES-Manufacturing Sample, replacing the CMIE macro proxies with one-wave-lagged BIES survey aggregates:

$$\text{INDUSTRY_SWITCH}_{i,t} = \sum_k \beta_k\,\text{BIES_LAG}_{k,t-1} + X'_{i,t}\gamma + \lambda_t + \alpha_i + \varepsilon_{i,t} \tag{3}$$

where $\text{BIES_LAG}_{k,t-1} \in \{\text{L1_q1_net_balance}_{t-1}, \text{L1_q2_net_balance}_{t-1}, \text{L1_q3_wtd_mean}_{t-1}, \text{L1_q4_wtd_mean}_{t-1}, \text{L1_q5_wtd_mean}_{t-1}\}$ are the lagged BIES profit-margin balance, sales-level balance, backward-looking cost change, forward-looking cost expectation, and CPI inflation expectation, respectively. Because BIES regressors vary only at the wave level (not at the individual or industry level), broad-industry fixed effects and industry-wave trends are omitted from Equation (3) to avoid collinearity with the wave fixed effects λ_t; individual fixed effects α_i are retained. For direct comparability, I estimate an analogous specification using the CMIE macro proxies (`L_CREDIT_INDUSTRY`, `L_IIP_INDUSTRY`, `L_WPI_INPUT`, `L_INF_CPI`) on the *same* BIES-Manufacturing Sample and wave set, so that any difference in explanatory power between the CMIE-proxy and BIES-direct columns of Table 5 reflects the information content of the regressor rather than a difference in sample composition. If the realised-conditions (CMIE) coefficients are significant while

the BIES-direct coefficients are not, this indicates that industry switching in the manufacturing subsample responds to realised macro conditions rather than to firms' own elicited expectations.

3.3 Leads-and-lags Specification: A test for pre-trends and reverse causality

A lagged coefficient being significant in Equation (2) is consistent with a causal, backward-looking channel running from realised macro conditions to industry switching, but it is equally consistent with reverse causality (industry switching itself shifting the industry-matched macro series) or with a pre-existing trend that happens to be correlated with the lag. A standard way to probe this is to add the one-wave-*led* value of each macro regressor alongside its lag: if switching behaviour today is not actually anticipating tomorrow's macro conditions, the led terms should be jointly indistinguishable from zero, which supports a causal reading of the lagged coefficients. I estimate,

$$\text{INDUSTRY_SWITCH}_{i,t} = \sum_k \beta_k \, X_{k,i,t-1} + \sum_k \phi_k \, X_{k,i,t+1} + X'_{i,t}\gamma + \delta_{j(i)} + \delta_{j(i)} \times \text{WAVE_NO} + \lambda_t + \alpha_i + \varepsilon_{i,t} \tag{4}$$

where $X_{k,i,t-1} \in$ L_INF_CPI, L_WPI_INPUT, L_CREDIT_INDUSTRY} are the same one-wave lags as in Equation (2), and $X_{k,i,t-1}$ are the corresponding one-wave-leads. The β_k retain their causal interpretation from Section **??**; the ϕ_k are the placebo coefficients with no causal meaning of their own and their role is more diagnostic. I estimate the Equation (4) on the main sample (Table 4 and for direct comparability with the BIES-Manufacturing subsample for robustness checks (Table 6), in both cases with individual and wave fixed effects and standard errors clustered at the individual level.

4 Results

In the one-way specification (Table 2), industry-matched IIP growth is positively and significantly associated with industry switching ($\hat{\beta} = 0.033$, $p < 0.01$): a one-point increase in lagged industry output growth is associated with a 3.3 percentage-point increase in the probability of switching industry in the current wave. This is consistent with the idea that expanding industries generate more job-to-job movement (both into and possibly out of the industry, since `INDUSTRY_SWITCH` does not distinguish direction), rather than with the Set A downward-mobility hypothesis specifically, which predicts a *negative* relationship between IIP growth and downward mobility - a hypothesis that requires the directional (`UPWARD_MOB/DOWNWARD_MOB`) outcomes rather than the undirected switch indicator used here. The coefficients on lagged CPI inflation, WPI input-cost inflation, and industry credit growth are small in magnitude and not statistically distinguishable from zero in the one-way specification.

Table 2: One-Way Fixed-Effects LPM: Industry Switching on Lagged CMIE Macro Variables

	(1) 1-Period Lag
CPI inflation, YoY %, 1-wave lag	-0.058
	(0.066)
WPI input-cost inflation, YoY %, 1-wave lag	-0.000
	(0.000)
Industry-matched IIP growth, YoY %, 1-wave lag	0.035***
	(0.012)
Industry-matched bank credit growth, YoY %, 1-wave lag	-0.001
	(0.001)
Observations	66,903
Within R^2	0.036

Standard errors in parentheses

Sample: Main Sample, earning members (EARNING_MEMBER=1) observed in >= 2 consecutive waves.

Sample period: CPHS waves 25-37 (approx. January 2022 - April 2026), the BIES coverage window; CMIE macro series and CPHS employment records are matched by wave.

Dependent variable: INDUSTRY_SWITCH (=1 if 2-digit NIC industry differs from the prior wave).

Estimator: linear probability model (LPM), individual fixed effects (xtreg, fe), one-way – Individual FE only, no wave FE.

Industry FE (BROAD_INDUSTRY_NAME, 4 categories: Agriculture, Manufacturing, Tangible Services, Intangible Services) and an industry x wave linear trend (BROAD_INDUSTRY_NAME continuous WAVE_NO) are included but not reported.

Individual controls (estimated, not reported): AGE_YRS (age, years), AGE2 (age squared), EDU_NUM (education, 0=illiterate to 9=postgraduate), MARRIED (currently married), BANK_ACC (has bank account), PF_COV (provident fund coverage), HEALTH_INS (health insurance), MOBILE_ACC (mobile phone access), RELATION_NUM (relationship-to-household-head dummies), WORK_PLACE_NUM (place-of-work dummies). Source: CMIE CPHS.

Regressors are 1-wave lags of CMIE Industry Outlook series (L_INF_CPI, L_WPI_INPUT, L_IIP_INDUSTRY, L_CREDIT_INDUSTRY); contemporaneous (unlagged) macro values are excluded by design.

Standard errors clustered at the individual level (MEM_PANEL_ID) in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Once wave fixed effects are added (Table 3), the IIP coefficient falls from 0.033 to 0.026 and loses statistical significance, while the point estimates on the other three regressors change little. This attenuation is expected: much of the wave-to-wave variation in IIP growth is common across industries (i.e. it reflects the aggregate business cycle documented in Figure 1), and once this common component is absorbed by λ_t, the remaining cross-industry variation used to identify β_3 in Equation (2) is considerably noisier. Overall, the Main Sample results provide only weak evidence that realised macro conditions predict undirected industry switching once aggregate wave effects are controlled for, which motivates the more targeted BIES-manufacturing comparison in Sections 5.1–5.2.

Table 3: Two-Way Fixed-Effects LPM: Industry Switching on Lagged CMIE Macro Variables

	(1) 1-Period Lag
CPI inflation, YoY %, 1-wave lag	-0.006
	(0.008)
WPI input-cost inflation, YoY %, 1-wave lag	-0.000
	(0.000)
Industry-matched IIP growth, YoY %, 1-wave lag	0.033
	(0.026)
Industry-matched bank credit growth, YoY %, 1-wave lag	-0.001
	(0.001)
Wave Fixed Effects	Yes
Observations	66,903
Within R^2	0.036

Standard errors in parentheses

Sample: Main Sample, earning members observed in >= 2 consecutive waves; identical sample to Table 2.

Sample period: CPHS waves 25-37 (approx. January 2022 - April 2026), the BIES coverage window; CMIE macro series and CPHS employment records are matched by wave.

Dependent variable: INDUSTRY_SWITCH.

Estimator: LPM, individual and wave fixed effects (xtreg, fe, plus i.WAVE_NO), two-way.

Identification of the four macro coefficients comes from cross-industry variation in macro conditions within a wave, net of the industry-specific linear trend, since wave FE absorb any shock common to all industries in a wave.

Industry FE (BROAD_INDUSTRY_NAME, 4 categories: Agriculture, Manufacturing, Tangible Services, Intangible Services) and an industry x wave linear trend (BROAD_INDUSTRY_NAME continuous WAVE_NO) are included but not reported.

Individual controls (estimated, not reported): AGE_YRS (age, years), AGE2 (age squared), EDU_NUM (education, 0=illiterate to 9=postgraduate), MARRIED (currently married), BANK_ACC (has bank account), PF_COV (provident fund coverage), HEALTH_INS (health insurance), MOBILE_ACC (mobile phone access), RELATION_NUM (relationship-to-household-head dummies), WORK_PLACE_NUM (place-of-work dummies). Source: CMIE CPHS.

Regressors are 1-wave lags of CMIE Industry Outlook series; contemporaneous values excluded.

Standard errors clustered at the individual level (MEM_PANEL_ID) in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 4 reports the leads-and-lags test of Equation (4) on the main sample. The lagged terms remain broadly consistent with the two-way FE results above: lagged CPI inflation is now positively significant (0.027), lagged WPI input-cost inflation and lagged IIP growth are negatively significant (-0.06 and -0.016 respectively) once the leads are included and the sample is reconstituted around the lead-lag window. More importantly for identification, all four *led* terms are statistically significant — F.CPI, F.WPI, F.IIP, and F.Credit. Because these placebo terms should be indistinguishable from zero under a purely backward-looking, causal reading of the lagged coefficients, this is evidence against a clean causal interpretation in the Main Sample: current industry switching behaviour is not independent of *future* realised macro conditions, consistent with either reverse causality, anticipatory switching ahead of macro conditions that were in fact to some extent forceable, or a pre-existing trend correlated with both. This qualifies the Main Sample findings in Sections 4 as descriptive robust correlations rather than a causal-identified adaptive-expectations channel.

Table 4: Leads-and-Lags Test: Industry Switching on Lagged and Led CMIE Macro Variables (Two-Way FE)

	(1) Lag + Lead
L.CPI inflation, YoY %	0.027**
	(0.011)
L.WPI input-cost inflation, YoY %	-0.006***
	(0.002)
L.Industry-matched IIP growth, YoY %	-0.016**
	(0.006)
oL.Industry-matched bank credit growth, YoY %	0.000
	(.)
F.CPI inflation, YoY %	-0.120**
	(0.050)
F.WPI input-cost inflation, YoY %	0.014**
	(0.006)
F.Industry-matched IIP growth, YoY %	0.098***
	(0.036)
F.Industry-matched bank credit growth, YoY %	-0.061***
	(0.023)
Controls	Yes
Industry FE	Yes
Industry x Wave Trend	Yes
Relationship-to-HOH FE	Yes
Place-of-Work FE	Yes
Wave Fixed Effects	Yes
Observations	132,798
Within R^2	0.018

Standard errors in parentheses

Sample: Main Sample, earning members observed in ¿=2 consecutive waves; identical sample to Table 2.

Sample period: CPHS waves 25-37 (approx. January 2022 - April 2026), the BIES coverage window; CMIE macro series and CPHS employment records are matched by wave.

Dependent variable: INDUSTRY_SWITCH.

Estimator: LPM, individual and wave fixed effects (xtreg, fe, plus i.WAVE_NO), two-way.

Regressors include both 1-wave lags (L1.) and 1-wave leads (F1.) of contemporaneous CMIE Industry Outlook series to test for pre-trends and dynamic impacts.

Industry FE (BROAD_INDUSTRY_NAME, 4 categories: Agriculture, Manufacturing, Tangible Services, Intangible Services) and an industry x wave linear trend (BROAD_INDUSTRY_NAME continuous WAVE_NO) are included but not reported.

Individual controls (estimated, not reported): AGE_YRS (age, years), AGE2 (age squared), EDU_NUM (education, 0=illiterate to 9=postgraduate), MARRIED (currently married), BANK_ACC (has bank account), PF_COV (provident fund coverage), HEALTH_INS (health insurance), MOBILE_ACC (mobile phone access), RELATION_NUM (relationship-to-household-head dummies), WORK_PLACE_NUM (place-of-work dummies). Source: CMIE CPHS.

Standard errors clustered at the individual level (MEM_PANEL_ID) in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

5 Robustness Check

5.1 BIES Direct Expectations

The Main Sample results in Section 4 use lagged *realised* macro conditions as adaptive-expectations proxies, since individual-level expectations are not observed for the full CPHS panel. BIES, however, directly elicits firm-level cost and inflation expectations for a manufacturing subsample of CPHS respondents, allowing a sharper test of whether *directly elicited* expectations matter for labour reallocation. This section restricts the sample to the BIES-Manufacturing Sample defined in Section **??** - CPHS manufacturing workers (NIC 10–33) in waves with BIES coverage, observed in at least two consecutive waves - and reports summary statistics before turning to the direct CMIE-versus-BIES model comparison in Section 5.2.

Table 1 shows that, on average, firms in the BIES-Manufacturing Sample report profit margins below normal (`L1_q1_net_balance` mean $= -0.412$) and sales levels below normal (`L1_q2_net_balance` mean $= -0.404$) relative to their own stated baseline, alongside a mean backward-looking cost-change assessment (`L1_q3_wtd_mean`) of 5.2%, a mean forward-looking cost expectation of 4.8%, and a mean CPI inflation expectation of 4.7% - both expectation measures somewhat below the realised cost-change assessment, consistent with firms expecting some moderation in cost pressure going forward. The BIES-Manufacturing industry-switch rate (39.2%) and lagged CMIE macro proxy means broadly mirror the Main Sample (Table 1), restricted to manufacturing waves with BIES coverage.

5.2 BIES VS. CMIE Direct Comparisons

Table 5 implements the head-to-head comparison of Equation (3) against the analogous CMIE-proxy specification, both estimated on the identical BIES-Manufacturing Sample and wave set with individual and wave fixed effects.

Column (1) uses the lagged CMIE macro proxies. Lagged WPI input-cost inflation is negatively and significantly associated with industry switching ($\hat{\beta} = -0.001$, $p < 0.05$), while lagged CPI inflation, industry-matched IIP growth, and industry-matched credit growth are not statistically significant, though the IIP coefficient is comparatively large in magnitude (0.152). Column (2) replaces these with the lagged BIES-elicited expectations: none of the five BIES regressors - the profit-margin balance, sales-level balance, backward-looking cost assessment, forward-looking cost expectation, or CPI inflation expectation - is statistically distinguishable from zero, despite a substantially larger sample (30,064 vs. 7,104 observations, reflecting greater non-missingness of the wave-level BIES aggregates relative to the industry-matched CMIE series). The two specifications also differ sharply in fit: the CMIE-proxy model explains considerably more within-individual variation (within $R^2 = 0.066$) than the BIES-direct model (within $R^2 = 0.005$).

Because a Set A (realised-conditions) coefficient is significant in column (1) while none of the Set B (BIES-direct) coefficients are significant in column (2), the evidence in this manufacturing subsample favours the conclusion that industry-switching behaviour responds to *realised* cost conditions - specifically wholesale input-cost inflation - rather than to firms' own directly elicited cost and inflation expectations. This runs counter to the framing implicit in the paper's research question, which anticipated an independent expectations channel; instead, the results suggest that, at least for the undirected industry-switching margin studied here, backward-looking realised shocks dominate forward-looking subjective expectations as a predictor of labour reallocation.

Table 5: Industry Switching: BIES Direct Expectations vs. CMIE Macro Proxies (Two-Way FE)

	(1) CMIE Proxies	(2) BIES Expectations
1-period lag: Credit industry growth	0.005	
	(0.006)	
1-period lag: industry IIP growth	0.163	
	(0.103)	
1-period lag: WPI input cost inflation	-0.001***	
	(0.000)	
1-period lag: CPI inflation	0.061	
	(0.052)	
BIES Q1: profit-margin net balance vs. normal, 1-wave lag (pp)		0.012
		(0.040)
BIES Q2: sales-level net balance vs. normal, 1-wave lag (pp)		0.013
		(0.143)
BIES Q3: backward-looking cost change, weighted mean, 1-wave lag (%)		-0.034
		(0.066)
BIES Q4: forward-looking cost expectation, weighted mean, 1-wave lag (%)		0.108
		(0.119)
BIES Q5: CPI inflation expectation, weighted mean, 1-wave lag (%)		0.056*
		(0.032)
Wave Fixed Effects	Yes	Yes
Observations	7,255	30,753
Within R^2	0.099	0.008

Standard errors in parentheses
Sample: BIES-Manufacturing Sample – CPHS manufacturing workers (NIC divisions 10-33) in waves with BIES coverage (has_bies=1), earning members observed in >= 2 consecutive waves.
Sample period: CPHS waves 25-37 (approx. January 2022 - April 2026), the BIES coverage window; CMIE macro series and CPHS employment records are matched by wave.
Dependent variable: INDUSTRY_SWITCH; identical outcome and identical sample/wave set in both columns, so any difference in fit reflects the information content of the regressor set rather than sample composition.
Column (1) regressors: 1-wave lags of the CMIE macro proxies (industry-matched credit growth, IIP growth, WPI input-cost inflation, CPI inflation).
Column (2) regressors: 1-wave lags of the five BIES-elicited firm expectation measures (Q1 margin balance, Q2 sales balance, Q3 backward cost change, Q4 forward cost expectation, Q5 CPI expectation); BIES regressors vary only at the wave level in this specification (see Section 6.6 for the industry-level alternative).
Estimator: LPM, individual and wave fixed effects (xtreg, fe, plus i.WAVE_NO).
Industry FE / industry-wave trend omitted: sample is single-industry (manufacturing) by construction, so these terms are not identified.
Individual controls (estimated, not reported): AGE_YRS (age, years), AGE2 (age squared), EDU_NUM (education, 0=illiterate to 9=postgraduate), MARRIED (currently married), BANK_ACC (has bank account), PF_COV (provident fund coverage), HEALTH_INS (health insurance), MOBILE_ACC (mobile phone access), RELATION_NUM (relationship-to-household-head dummies), WORK_PLACE_NUM (place-of-work dummies). Source: CMIE CPHS.
Standard errors clustered at the individual level (MEM_PANEL_ID) in parentheses.
* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Table 6: Leads-and-Lags Test: Industry Switching on Lagged and Led CMIE Macro Proxies, BIES-Manufacturing Sample (Two-Way FE)

	(1) Lag + Lead
L.CPI All Items YoY %	0.018
	(0.023)
L.WPI input cost matched to worker industry	-0.001
	(0.010)
L.IIP growth matched to worker NIC industry	0.006
	(0.019)
oL.Bank credit growth matched to worker industry	0.000
	(.)
F.CPI All Items YoY %	-0.032
	(0.025)
F.WPI input cost matched to worker industry	-0.009
	(0.009)
F.IIP growth matched to worker NIC industry	-0.043*
	(0.023)
F.Bank credit growth matched to worker industry	0.016
	(0.010)
Controls	Yes
Relationship-to-HOH FE	Yes
Place-of-Work FE	Yes
Wave Fixed Effects	Yes
Observations	10,671
Within R^2	0.008

Standard errors in parentheses

Sample: BIES-Manufacturing Sample – CPHS manufacturing workers (NIC divisions 10-33) in waves with BIES coverage (has_bies=1), earning members observed in >= 2 consecutive waves.

Sample period: CPHS waves 25-37 (approx. January 2022 - April 2026), the BIES coverage window; CMIE macro series and CPHS employment records are matched by wave.

Dependent variable: INDUSTRY_SWITCH.

Estimator: LPM, individual and wave fixed effects (xtreg, fe, plus i.WAVE_NO), two-way.

Regressors include both 1-wave lags (L1.) and 1-wave leads (F1.) of CMIE macro proxies to test for pre-trends in the BIES sub-sample.

Individual controls (estimated, not reported): AGE_YRS (age, years), AGE2 (age squared), EDU_NUM (education, 0=illiterate to 9=postgraduate), MARRIED (currently married), BANK_ACC (has bank account), PF_COV (provident fund coverage), HEALTH_INS (health insurance), MOBILE_ACC (mobile phone access), RELATION_NUM (relationship-to-household-head dummies), WORK_PLACE_NUM (place-of-work dummies). Source: CMIE CPHS.

Standard errors clustered at the individual level (MEM_PANEL_ID) in parentheses.

* $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$

Section 3.3 flagged that the Main Sample lagged coefficients fail a leads-and-lags pre-trend test, so I repeat that test here on the CMIE macro proxies within the BIES-Manufacturing subsample and wave set, to see whether the WPI result of column (1) above survives the same scrutiny.

Table 6 shows a markedly cleaner pattern than the Main Sample counterpart (Table 4). None of the four lagged terms is statistically significant in the smaller sample, and of the four led terms, only F.IIP is weakly significant ($\hat{\phi} = -0.043$, $p < 0.10$); F.CPI, F.WPI, and F.Credit are all statistically indistinguishable from zero. Because the placebo (led) terms are largely null here, this leads-and-lags test does not turn up the same reverse causality or pre-trend concern that it does in the Main Sample, which leads more credibility to a causal reading of the realised WPI input-cost result reported in column (1) of Table 5, though the loss of significance on the WPI *lag* itself once leads are added, together with the smaller cell sizes underlying this subsample, means this should be seen as a supportive causal confirmation rather than definitive.

These finding should be interpreted with the caveats that (i) the BIES-Manufacturing Sample is substantially smaller and more volatile than the Main Sample, and (ii) BIES regressors vary only at the wave level and so are identified off considerably less variation than the industry-matched CMIE series.

6 Conclusion

This paper asked whether inflationary expectations affect labour mobility and industry switching among Indian workers, using a merged CMIE CPHS–BIES panel spanning waves 25–39 (January 2022–April 2026). Using individual and industry (or wave) fixed-effects linear probability models of industry switching on lagged realised macro conditions, the Main Sample results (Section 4) show only limited evidence that realised macro conditions predict industry switching once aggregate wave effects are absorbed: industry-matched IIP growth is significantly positive in a one-way specification but loses significance once wave fixed effects are added. Restricting to a manufacturing subsample with BIES coverage and directly comparing lagged CMIE macro proxies against lagged BIES-elicited firm expectations (Sections 5.1–5.2) sharpens this picture: lagged WPI input-cost inflation significantly predicts industry switching, while none of the five BIES expectation measures do, despite a larger estimation sample. Taken together, these results suggest that, for the undirected industry-switching margin examined here, labour reallocation among Indian workers tracks *realised* cost shocks more closely than it tracks firms' own *elicited* expectations of future costs and inflation - a finding that qualifies, rather than confirms, the expectations-channel hypothesis motivating this paper. A leads-and-lags placebo test (Section 3.3) qualifies this further, in the Main Sample, where current switching is predicted not only by lagged macro conditions but also by their one-wave-*led* values. Therefore, the Main Sample lag coefficients are to be seens as robust descriptive correlations rather than causally identified effects. The same placebo test on the BIES-Manufacturing subsample, where the led terms are largely insignificant giving somewhat more support to a causal reading of the realised WPI input-cost result that anchors the paper's central claim.

Returning to the question raised in Section 2.1: what does this exercise add beyond Akarsu et al. (2025)[1]'s causal finding that experimentally induced expectations revisions move Turkish firms' pricing and hiring? The contribution is not causal test of the same mechanism in a new country, since the BIES-CPHS design here has no exogenous source of variation in expectations. Rather, the finding that realised conditions (WPI input-cost inflation, and more weakly, industry output growth) predict worker mobility while wave aggregates elicited expectations do not, which is consistent with the structural story developed in 2.1: that in a labour market where dominant adjustment margin is movement across formal, informal, and casual employment tiers rather than formal hiring and firing, workers and firms alike may reallocate in response to shocks that have already materialised rather than shocks they anticipate. An equally plausible alternative reading, which the present

design cannot rule out, is purely methodological: BIES expectations are observed only at the wave level and merged onto workers by industry rather than by firm, so any true firm-level expectations effect of the kind Akarsu et al. identify could be attenuated to statistical insignificance by this aggregation and mismatch, independent of whether the underlying mechanism is weaker in India. Adjudicating between these two explanations - a genuinely weaker expectations channel versus an aggregation artefact - requires either a firm-to-worker matched BIES-CPHS linkage or a source of plausibly exogenous variation in Indian firms' expectations (for instance, differential regional or sectoral exposure to a common information shock, in the spirit of Akarsu et al.'s randomised design); both are natural next steps and are noted among the limitations below.

Several limitations point to directions for further work. First, the outcome variable used throughout is the undirected industry-switch indicator. Second, the BIES-Manufacturing Sample is small relative to the Main Sample and BIES regressors are identified only off wave-level variation, which limits statistical power relative to the industry-matched CMIE proxies; a longer BIES panel or firm-level (rather than wave-aggregated) BIES linkage would sharpen this comparison. Third, the heterogeneity analysis by gender and rural/urban location motivated by Gautier et al. (2025)[5] and Chen et al. (2003)[3] in Section 2, and the financial-inclusion buffer variables (bank account, provident fund, health insurance, mobile access) motivated by Kumari et al. (2025)[8], are constructed in the Appendix variable set (Table 7) but not yet incorporated into the reported specifications. Addressing these three extensions is left for the next stage of this research.

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7 Appendix

7.1 Variables

Table 7: Variable Definitions and Sources

Variable	Definition	Source
MEM_PANEL_ID	Unique individual panel identifier	CMIE CPHS
WAVE_NO	Survey wave number (panel time variable)	CMIE CPHS
OCC_SWITCH	Binary: 1 if individual switches primary occupation (NCO group) between consecutive waves	CMIE CPHS
IND_SWITCH	Binary: 1 if individual switches 2-digit NIC industry between consecutive waves	CMIE CPHS
UPWARD_MOB	Binary: 1 if individual moves to a higher occupational tier (NAT_OCC_TIER increases)	CMIE CPHS
DOWNWARD_MOB	Binary: 1 if individual moves to a lower occupational tier (NAT_OCC_TIER decreases)	CMIE CPHS
NIC_UPWARD_MOB	Binary: 1 if UPWARD_MOB=1 and IND_SWITCH=1	CMIE CPHS
NIC_DOWNWARD_MOB	Binary: 1 if DOWNWARD_MOB=1 and IND_SWITCH=1	CMIE CPHS
NCO_UPWARD_MOB	Binary: 1 if UPWARD_MOB=1 and OCC_SWITCH=1	CMIE CPHS
NCO_DOWNWARD_MOB	Binary: 1 if DOWNWARD_MOB=1 and OCC_SWITCH=1	CMIE CPHS
FORMAL	Binary: 1 if individual is in formal (salaried) employment	CMIE CPHS
INFORMAL	Binary: 1 if individual is in informal employment	CMIE CPHS
CASUAL	Binary: 1 if individual is in casual wage employment	CMIE CPHS
SELF_EMP	Binary: 1 if individual is self-employed	CMIE CPHS
EARNING_MEMBER	Binary: 1 if individual is currently employed (earning)	CMIE CPHS
EMP_TYPE_EXTENDED	Encoded categorical: extended employment and activity classification	CMIE CPHS
NAT_OCC_TIER	Ordered 1–3 tier classification of occupation quality	CMIE CPHS
EMP_QUALITY	Ordered 1–4 quality score: 1=casual, 4=permanent salaried	CMIE CPHS
INF_CPI	CPI All-Items Year-on-Year inflation (%)	CMIE Industry Outlook
INF_FOOD	CPI Food Year-on-Year inflation (%)	CMIE Industry Outlook

Variable	Definition	Source
INF_CORE	CPI Core (ex-food, ex-fuel) Year-on-Year inflation (%)	CMIE Industry Outlook
WPI_INPUT	WPI Input Cost Index Year-on-Year growth (%)	CMIE Industry Outlook
IIP_SECTOR	Index of Industrial Production, sector-matched YoY growth (%)	CMIE Industry Outlook
CREDIT_SECTOR	Sectoral bank credit growth Year-on-Year (%)	CMIE Industry Outlook
L_INF_CPI	1-wave lag of INF_CPI; adaptive-expectations proxy	CMIE Industry Outlook
L2_INF_CPI	2-wave lag of INF_CPI	CMIE Industry Outlook
L_WPI_INPUT	1-wave lag of WPI_INPUT	CMIE Industry Outlook
L2_WPI_INPUT	2-wave lag of WPI_INPUT	CMIE Industry Outlook
L_IIP_SECTOR	1-wave lag of IIP_SECTOR	CMIE Industry Outlook
L2_IIP_SECTOR	2-wave lag of IIP_SECTOR	CMIE Industry Outlook
L_CREDIT_SECTOR	1-wave lag of CREDIT_SECTOR	CMIE Industry Outlook
L2_CREDIT_SECTOR	2-wave lag of CREDIT_SECTOR	CMIE Industry Outlook
q3_wtd_mean	BIES Q3: wave-avg weighted mean current cost change (%)	BIES (IIMA)
q4_wtd_mean	BIES Q4: wave-avg weighted mean forward cost expectation (%)	BIES (IIMA)
q5_wtd_mean	BIES Q5: wave-avg weighted mean CPI inflation expectation (%)	BIES (IIMA)
q1_net_balance	BIES Q1: profit margin net balance (pp above minus below normal)	BIES (IIMA)
q2_net_balance	BIES Q2: sales level net balance (pp above minus below normal)	BIES (IIMA)
q4_sd	BIES Q4: std dev of forward cost distribution (uncertainty measure)	BIES (IIMA)
L1_q4_wtd_mean	1-wave lag of q4_wtd_mean	BIES (IIMA)
L2_q4_wtd_mean	2-wave lag of q4_wtd_mean	BIES (IIMA)
L1_q5_wtd_mean	1-wave lag of q5_wtd_mean	BIES (IIMA)
has_bies	Binary: 1 if CPHS wave has matching BIES data	BIES merge
AGE_YRS	Age of individual in years	CMIE CPHS
AGE2	Age squared	CMIE CPHS
GENDER_BIN	Binary: 1 if female, 0 if male	CMIE CPHS
REGION	Binary: 1 if urban, 0 if rural	CMIE CPHS
EDU_NUM	Numeric education level (0=illiterate, 9=postgraduate)	CMIE CPHS
MARRIED	Binary: 1 if currently married	CMIE CPHS
BANK_ACC	Binary: 1 if individual has a bank account	CMIE CPHS
PF_COV	Binary: 1 if covered by provident fund	CMIE CPHS
HEALTH_INS	Binary: 1 if covered by health insurance	CMIE CPHS

Variable	Definition	Source
MOBILE_ACC	Binary: 1 if has mobile phone access	CMIE CPHS
HOH	Binary: 1 if individual is head of household	CMIE CPHS
MIGRANT	Binary: 1 if interstate migrant	CMIE CPHS
HOME_BASED	Binary: 1 if works from home	CMIE CPHS
RELATION_NUM	Numeric encoding of relationship to head of household	CMIE CPHS
WORK_PLACE_NUM	Numeric encoding of place of work	CMIE CPHS

7.2 BIES Substitutes: Macroeconomic Variables

Table 8 – Continued from previous page

BIES Question Reference	Exact CMIE Master Indicator Name	Description & Econometric Transmission Channel
Q1: How do your current PROFIT MARGINS compare with "normal" times?	SCBs Credit Non-Food by Economic Activity (Focusing on the Services & Industry master branches)	Measures the growth of institutional lending to specific sectors (e.g., Trade, Transport, IT, Textiles). If YoY bank credit growth to a worker's industry drops, it signals a liquidity squeeze and compressed operational margins, which empirically forces firms to freeze hiring or cut wages.
Q2: How do your current sales levels compare with SALES LEVELS during what you consider to be "normal" times?	Index of Industrial Production (IIP): 2-Digit Manufacturing Industry Activity	Measures the physical volume velocity of manufacturing divisions mapped explicitly to 2-digit NIC codes (e.g., Division 10: Food Products, Division 13: Textiles). A contraction in YoY output tracks the real-sector slowdown driving labour displacement.
Q3 & Q4: Looking back/Projecting ahead, how do your current COSTS PER UNIT compare with this time last year / next 12 months?	WPI: Manufactured Products (All Sub-sectors); WPI: Primary Articles (Commodity/Extractive)	Contains separate columns for dozens of distinct industrial supply chains (e.g., WPI: Manufacture of Chemicals, WPI: Basic Metals). Represents the primary supply-side cost shock vector affecting a firm's willingness to sustain labour costs.
Q5: Projecting ahead... please assign a percent likelihood (probability) to the CPI inflation over the next 12 months.	CPI: Combined Group-wise / Sectoral Indices	Breaks down consumer inflation into sector-facing retail indices (e.g., CPI: Clothing & Footwear, CPI: Fuel & Light, CPI: Housing). Measures how much of an industry's wholesale cost shock is passed down to consumer retail prices, mapping nominal wage pressures.

Continued on next page...

7.3 BIES Interval Mapping

Table 9: BIES Survey Interval Mapping and Midpoints

Question	Original Survey Interval	Assigned Midpoint (M)
Q3/Q4: Cost Down	$< -1\%$	-2.0%
Q3/Q4: Cost Unchanged	-1% to 1%	0.0%
Q3/Q4: Cost Up Somewhat	1.1% to 3%	2.0%
Q3/Q4: Cost Up Moderately	3.1% to 6%	4.5%
Q3/Q4: Cost Up Significantly	6.1% to 10%	8.0%
Q3/Q4: Cost Up Very Significantly	$> 10\%$	12.0%
Q5: CPI below -1%	$< -1\%$	-1.5%
Q5: CPI 1-2%	1% to 2%	1.5%
Q5: CPI 2-3%	2.1% to 3%	2.55%
Q5: CPI 3-4%	3.1% to 4%	3.55%
Q5: CPI 4-5%	4.1% to 5%	4.55%
Q5: CPI 5-6%	5.1% to 6%	5.55%
Q5: CPI 6-7%	6.1% to 7%	6.55%
Q5: CPI 7-8%	7.1% to 8%	7.55%
Q5: CPI above 8%	$> 8\%$	9.0%

Source: BIES (IIMA Misra Centre) survey documentation, as applied to the CPHS-BIES merged panel.

7.4 Control Variables

Table 10: Control Variables: Economic and Structural Description

Variable	Description
`AGE_YRS`, `AGE2`	Age in years and its square. Captures lifecycle position and human-capital stock.
`EDU_NUM`	Ordered numeric education level. Higher education is expected to reduce distress-driven mobility.
`GENDER_BIN`	Binary: 1 if female. Dropped in FE estimators but retained in split-sample heterogeneity.
`REGION`	Binary: 1 if urban. Treated symmetrically with `GENDER_BIN`.
`MARRIED`	Binary: 1 if currently married. Predicts labour-supply and occupational-mobility decisions.
`BANK_ACC`	Binary: 1 if the individual holds a bank account. Proxies formal-sector liquidity buffers.
`PF_COV`	Binary: 1 if covered by a provident fund. Indicates formal employment attachment.
`HEALTH_INS`	Binary: 1 if covered by health insurance. Proxies formal employment and welfare-scheme access.
`MOBILE_ACC`	Binary: 1 if the individual has mobile-phone access. Captures digital connectivity.
`RELATION_NUM`	Numeric encoding of individual's relationship to household head.
`WORK_PLACE_NUM`	Numeric encoding of work location. Proxies informality and fragile structural arrangements.

7.5 Estimation notes

- **Operationalisation of macroeconomic expectations:** Since firm- and worker-level expectations survey data are not available at the required individual-wave frequency for the full CPHS sample, macroeconomic expectations are operationalised throughout using the adaptive-expectations framework of Equation (1). The one-wave (L1) and two-wave (L2) lags of each macro regressor serve as empirical proxies for the macroeconomic expectations.
- **Fixed Effects Setup:** All FE models absorb `MEM_PANEL_ID` as the individual fixed effect and include wave fixed effects via `i.WAVE_NO` where applicable. The within-estimator differences out all time-invariant individual characteristics.
- **Standard Errors:** Standard errors are clustered at the individual level (`vce(cluster MEM_PANEL_ID)`) in all specifications across all three analysis tracks.
- **BIES Wave-level Regressors:** BIES regressors vary only at the wave level. Wave fixed effects are therefore omitted from BIES-track regressions (Sections 6-7) to avoid perfect collinearity.
- **BIES Coverage Period:** The BIES coverage period corresponds to CPHS waves 25-39 (approximately January 2022 – April 2026).
- **Mobility Sample Constraints:** NIC- and NCO-interaction-based mobility outcomes are defined by the interaction equations of Section 3.2. The combined mobility tables (Tables 14-15) restrict the sample to `EARNING_MEMBER` $= 1$ and require at least two consecutive panel observations per individual.
- **Statistical significance:** * $p < 0.10$, ** $p < 0.05$, *** $p < 0.01$.